

IMO Training – Number Theory

Law of Quadratic Reciprocity

Miscellaneous Facts in Number Theory

Longtin Chan
longtinclt@gmail.com

25 November, 2023

Introduction

Unless otherwise specified, we shall focus on integers. Furthermore, let p be an odd prime.

Definition. Let $(a, m) = 1$. Then a is a *quadratic residue mod m* if $\exists x \in \mathbb{Z}$ such that $x^2 \equiv a \pmod{m}$. Otherwise, we call a is a *quadratic nonresidue mod m* .

Definition. The *Legendre's Symbol* $\left(\frac{a}{p}\right)$ is defined by

$$\left(\frac{a}{p}\right) = \begin{cases} +1 & \text{if } a \text{ is a quadratic residue mod } p \\ -1 & \text{if } a \text{ is a quadratic nonresidue mod } p \\ 0 & \text{if } p \mid a \end{cases}$$

Theorem. (Quadratic Reciprocity Law) Let p, q are distinct odd primes, then

$$\left(\frac{p}{q}\right) \left(\frac{q}{p}\right) = (-1)^{\left(\frac{p-1}{2}\right)\left(\frac{q-1}{2}\right)}.$$

Theorem. (Fermat's Two Square Theorem) Any prime p of the form $4k + 1$ is a sum of two squares.

Theorem. (Fermat's Last Theorem) If $x, y, z, n \in \mathbb{Z}$, $n \geq 3$ and $x^n + y^n = z^n$, then $xyz = 0$.

Theorem. (Dirichlet) If $(a, b) = 1$, then there are infinitely many primes of the form $an + b$.

Theorem. (Catalan) The only integral solution to the equation $a^b - c^d = 1$ with $a, b, c, d > 1$ is $(a, b, c, d) = (3, 2, 2, 3)$.

Theorem. (Fibonacci Power) 1, 8, and 144 are the only perfect powers in the Fibonacci sequence.

Proposition. (Euler's Criterion) For all a , $\left(\frac{a}{p}\right) \equiv a^{\frac{p-1}{2}} \pmod{p}$.

Proposition. $\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right) \left(\frac{b}{p}\right)$

Example. Find $\left(\frac{-1}{p}\right)$ and $\left(\frac{2}{p}\right)$.

Example. Evaluate the following:

1. $\left(\frac{102}{1021}\right)$

2. $\left(\frac{778}{9931}\right)$

3. $\left(\frac{973}{2711}\right)$

4. $\left(\frac{252000}{2591}\right)$

Example. Prove that -3 is not a primitive root of any prime of the form $4^n + 3$.

Example. Solve $x^2 + 3xy - 2y^2 = 122$ for integers.

Example. Show that there are infinitely many primes of the form $4k + 1$.

Example. Show that for any $a, b \in \mathbb{N}$, $a^2 + b$ and $b^2 + a$ cannot be both perfect squares.

Example. Find all $x, y \in \mathbb{Z}$ s.t. $x^5 = y^2 + 4$.

Example. Find all $x, y \in \mathbb{Z}$ s.t. $3x^2 + 1 = 4y^3$.

Exercises

1. (AIME 2001) How many positive integer multiples of 1001 can be expressed in the form $10^j - 10^i$, where i and j are integers and $0 \leq i < j \leq 99$?
2. (IMO 1978) Let m and n be natural numbers such that $1 \leq m < n$. In their decimal representations, the last three digits of 1978^m are equal, respectively, to the last three digits of 1978^n . Find m and n such that $m + n$ has its least value.
3. (IMO 1990) Find all positive integers $n > 1$ such that $\frac{2^n + 1}{n^2}$ is an integer.
4. (Bulgaria 1996) Find all primes p, q s.t. $pq \mid (5^p - 2^p)(5^q - 2^q)$.
5. (USA TST 2003) Find all ordered prime triples (p, q, r) s.t. $p \mid q^r + 1$, $q \mid r^p + 1$, $r \mid p^q + 1$.
6. (China 2009) Find all primes p, q s.t. $pq \mid 5^p + 5^q$.
7. (China 2005) Find all $x, y, z, w \in \mathbb{Z}_{\geq 0}$ s.t. $2^x 3^y - 5^z 7^w = 1$.
8. (Bulgaria 2006) Find all $t, x, y, z \in \mathbb{N}$ s.t. $2^t = 3^x 5^y + 7^z$.
9. Find all $x, y \in \mathbb{Z}$ s.t. $x^3 - y^3 = xy + 61$.
10. Find all $x, y \in \mathbb{Z}$ s.t. $x^2 - y! = 2001$.
11. For $m, n \in \mathbb{N}$, find the minimum value of $|12^m - 5^n|$.
12. Find all $x, y \in \mathbb{Z}$ s.t. $x^3 + y^4 = 19^{19}$.
13. Find all $k \in \mathbb{N}$ s.t. $\exists a, b \in \mathbb{N}$ with $a^2 + b^2 = kab$.
14. If $a, b \in \mathbb{N}$ and $b^2 + ab + 1 \mid a^2 + ab + 1$, prove that $a = b$.
15. Show that for $a, b, c \in \mathbb{N}$ with $a^2 + b^2 = c^2$, $\exists m, n \in \mathbb{Z}$ s.t. $(a, b, c) = (m^2 - n^2, 2mn, m^2 + n^2)$.
16. Find all $x, y \in \mathbb{Z}$ s.t. $x^2 - 2y^2 = 1$.
17. (Bulgaria 1998) Let $m, n \in \mathbb{N}$ such that $A = \frac{(m+3)^n + 1}{3m} \in \mathbb{Z}$. Prove that A is odd.
18. (IMO 1996) Let $a, b \in \mathbb{N}$ s.t. $15a + 16b, 16a - 15b$ are both perfect squares. Find the minimum value of the smaller square.